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(54) **ELECTRONIC DEVICE WITH IMPROVED DRIVING MECHANISM STRUCTURE**
(71) Applicant: **VIVO MOBILE COMMUNICATION CO., LTD.**, Guangdong (CN)
(72) Inventor: **Xiaojun Yi**, Guangdong (CN)
(73) Assignee: **VIVO MOBILE COMMUNICATION CO., LTD.**, Guangdong (CN)
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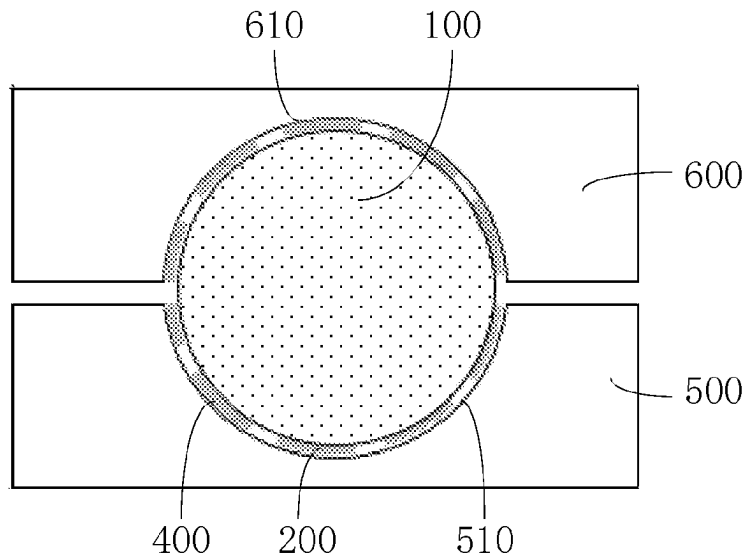
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Primary Examiner — Emily P Pham
(74) *Attorney, Agent, or Firm* — BROOKS KUSHMAN P.C.

(57) **ABSTRACT**
An electronic device is provided, including a shell, a driving mechanism and a driven module, wherein the shell is provided with an avoidance space communicating with an inner cavity of the shell; the driving mechanism includes at least two first field deformation structural components; the first field deformation structural components are arranged in the avoidance space; the adjacent two first field deformation structural components are distributed at intervals and energizing currents are opposite; the driven module is connected to the first field deformation structural components; and when the first field deformation structural components are in a power-up state, each of the first field deformation structural components drives the driven module to rotate through deformation.

11 Claims, 3 Drawing Sheets



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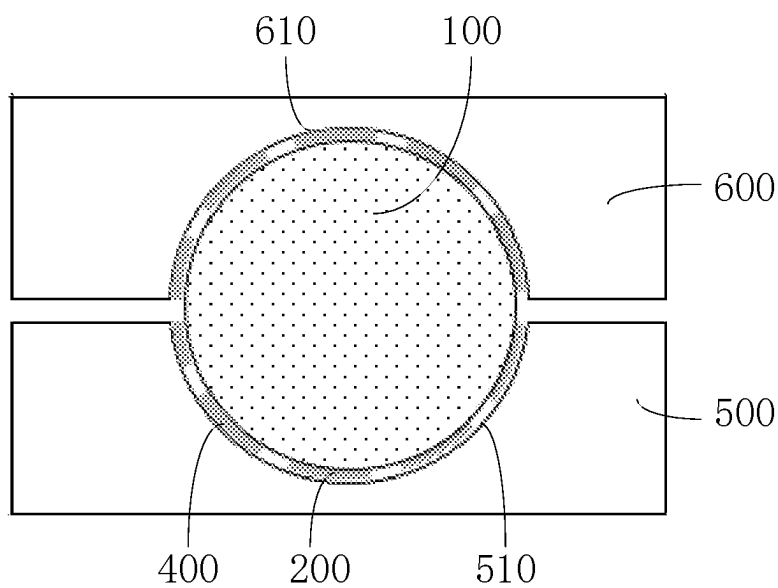


FIG. 1

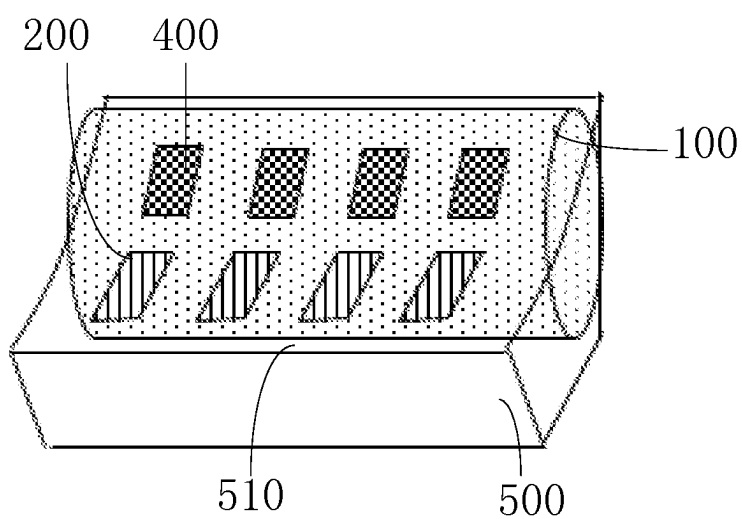


FIG. 2

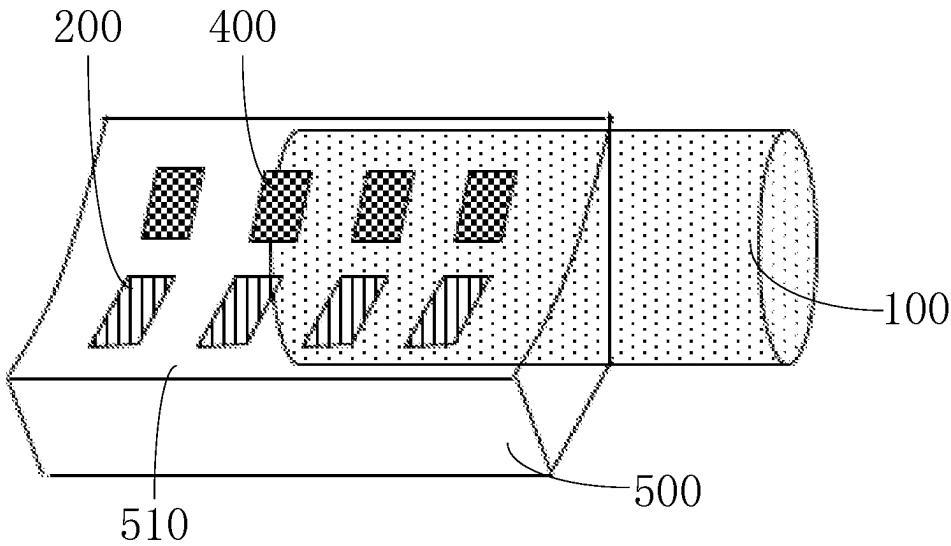


FIG. 3

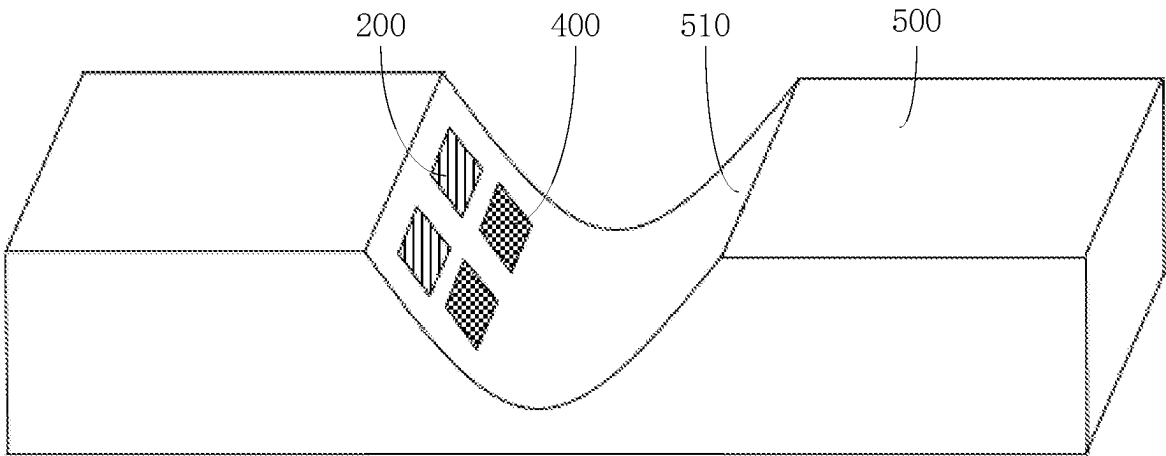


FIG. 4

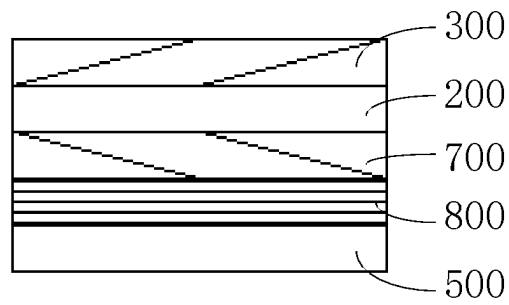


FIG. 5

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**ELECTRONIC DEVICE WITH IMPROVED
DRIVING MECHANISM STRUCTURE****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation application of International Application No. PCT/CN2020/121916 filed on Oct. 19, 2020, which claims priority to Chinese Patent Application No. 201911013990.1 filed on Oct. 23, 2019, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present disclosure relates to the technical field of communication, and in particular, to an electronic device.

BACKGROUND

With the progress of technologies and the development of electronic devices, users' demands for full-screen electronic devices are gradually increasing. With the development of the full-screen electronic devices, some driven modules are arranged in the electronic device; meanwhile, in order to make the driven modules realize corresponding functions, the electronic device further includes a driving mechanism, and the driving mechanism is used to drive the driven modules to realize corresponding functions.

Usually, the driving mechanism mainly includes a driving motor and a transmission device. However, since the driving motor and the transmission device have relatively complex structures, the volume is relatively large, so that the internal spatial arrangement of the electronic device is affected; meanwhile, the driving device is generally a metal material, so that the overall weight of the driving mechanism is relatively large, and a certain noise will be generated when the driving motor works, thereby affecting user experience.

SUMMARY

The present disclosure provides an electronic device.

An electronic device includes a shell, a driving mechanism and a driven module, wherein the shell is provided with an avoidance space communicating with an inner cavity of the shell; the driving mechanism includes at least two first field deformation structural components; the first field deformation structural components are arranged in the avoidance space; the adjacent two first field deformation structural components are distributed at intervals and energizing currents are opposite; the driven module is connected to the first field deformation structural components; and when the first field deformation structural components are in a power-up state, each of the first field deformation structural components drives the driven module to rotate through deformation.

The technical solutions used in the present disclosure can achieve the following beneficial effects:

The electronic device disclosed by the embodiments of the present disclosure utilizes vibration of a plurality of first field deformation structural components to form traveling wave propagated along the rotating direction of the driven module, so as to drive the driven module to rotate. Compared with existing driving mechanisms, in the embodiments disclosed by the present disclosure, the driving mechanism does not require a driving motor and a transmission device, so that the structure of the driving mechanism is relatively simple, the internal space of the electronic

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device occupied by the driving mechanism is relatively small, and the generated noise is relatively low.

BRIEF DESCRIPTION OF DRAWINGS

The accompanying drawings described herein are used to provide further understanding of the present disclosure and constitute a part of the present disclosure. The illustrative embodiments of the present disclosure and descriptions thereof are used to explain the present disclosure, and do not constitute an improper limitation to the present disclosure. In the accompanying drawings:

FIG. 1 is a structural schematic diagram of an electronic device according to an embodiment of the present disclosure;

FIG. 2 is a schematic diagram of a partial structure of a driven module of an electronic device according to an embodiment of the present disclosure in a retracted state;

FIG. 3 is a schematic diagram of a partial structure of a driven module of an electronic device according to an embodiment of the present disclosure in a stretched state; and

FIG. 4 and FIG. 5 are schematic diagrams of partial structures of an electronic device according to an embodiment of the present disclosure.

DESCRIPTION OF REFERENCE NUMERALS

- 100**—Driven module;
- 200**—First field deformation structural component;
- 300**—First elastic portion;
- 400**—Second field deformation structural component;
- 500**—Middle frame, **510**—First groove;
- 600**—Rear cover, **610**—Second groove;
- 700**—Circuit layer;
- 800**—Insulating layer.

DETAILED DESCRIPTION

To make the objectives, technical solutions, and advantages of the present disclosure clearer, the following clearly describes the technical solutions of the present disclosure with reference to specific embodiments of the present disclosure and the accompanying drawings. Apparently, the described embodiments are some rather than all of the embodiments of the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present disclosure without creative efforts shall fall within the protection scope of the present disclosure.

The following describes in detail the technical solutions disclosed in the embodiments of the present disclosure with reference to the accompanying drawings.

As shown in FIG. 1 to FIG. 5, the present disclosure discloses an electronic device. The disclosed electronic device includes a shell, a driving mechanism and a driven module **100**.

The shell is a peripheral component of the electronic device. The shell can provide a mounting position for other components of the electronic device. In the embodiments of the present disclosure, the shell is provided with an avoidance space communicating with an inner cavity of the shell.

The driving mechanism is connected to the driven module **100**. The driving mechanism can drive the driven module **100** to move. Specifically, the driving mechanism includes at least two first field deformation structural components **200**; the first field deformation structural components **200** are

arranged in the avoidance space; the adjacent two first field deformation structural components **200** are distributed at intervals and energizing currents are opposite; the driven module **100** is connected to the first field deformation structural components **200**; and the first field deformation structural components **200** can drive the driven module **100** to move through deformation.

The driven module **100** may include at least one of a camera assembly, a light supplementing module, a fingerprint identification module, a USB interface and a receiver. The embodiments of the present disclosure do not limit the specific types of the driven module **100**.

In the specific working process, the first field deformation structural components **200** are in a power-up state, each of the first field deformation structural components **200** is subjected to telescopic deformation, and any adjacent two first field deformation structural components **200** are subjected to alternatively telescopic deformation, so that the at least two first field deformation structural components **200** are arranged to form traveling wave propagated along the rotating direction of the driven module **100**, and the driven module **100** is driven to rotate. In this case, the driven module **100** may rotate to any direction, thereby avoiding the requirements of arranging a plurality of driven modules **100** to realize the work of the driven modules **100** in multiple directions. In the embodiments of the present disclosure, the process and the principle of forming traveling wave through telescopic deformation after at least two first field deformation structural components **200** are energized are well-known technologies, and will not be elaborated herein.

Of course, the shell and the display screen of the electronic device may be provided with light-transmitting areas, and the driven module **100** may work through the light-transmitting areas.

It can be known from the above working process that the electronic device disclosed by the embodiments of the present disclosure utilizes vibration of a plurality of first field deformation structural components **200** to form traveling wave propagated along the rotating direction of the driven module **100**, so as to drive the driven module **100** to rotate. Compared with existing driving mechanisms, in the embodiments of the present disclosure, the driving mechanism does not require a driving motor and a transmission device, so that the structure of the driving mechanism is relatively simple, the internal space of the electronic device occupied by the driving mechanism is relatively small, and the generated noise is relatively low.

In the electronic device disclosed by the embodiments of the present disclosure, in order to improve the driving effect of the first field deformation structural components **200** on the driven module **100**, the driving module may further include a first elastic portion **300**. Specifically, the first elastic portion **300** is arranged between the driven module **100** and the first field deformation structural components **200**. The first elastic portion **300** may deform along with the deformation of the first field deformation structural components **200**, and the first elastic portion **300** can enlarge the traveling wave formed by the first field deformation structural components **200**, so that the driving effect of the first field deformation structural components **200** on the driven module **100** is more obvious.

Of course, in order to prevent the work of the driven module **100** from being affected by the shell, in the specific working process, at least part of the driven module **100** may stretch out of the shell for work through the avoidance space. At the same time, after the work is completed, the driven

module **100** can retract in the shell through the avoidance space. Therefore, in order to make the driven module **100** retract in the shell or at least partially stretch out of the shell, the driving mechanism may further include at least two second field deformation structural components **400**. Similarly, the second field deformation structural components **400** may also be arranged in the avoidance space; and the adjacent two second field deformation structural components **400** are distributed at intervals and energizing currents are opposite.

In the specific working process, the second field deformation structural components **400** are in a power-up state, each of the second field deformation structural components **400** is subjected to telescopic deformation, and any adjacent two second field deformation structural components **400** are subjected to alternatively telescopic deformation, so that the at least two second field deformation structural components **400** are arranged to form traveling wave propagated along the telescoping direction of the driven module **100**, and the driven module **100** can be driven to retract the shell through the avoidance space or at least partially stretch out of the shell. Of course, in order to realize the movement of the driven module **100** in two directions, it is only necessary to change the direction of the energizing current of the second field deformation structural components **400**.

Of course, in this case, the first field deformation structural components **200** and the second field deformation structural components **400** may cooperate mutually so as to drive the driven module **100** to move. Specifically, the first field deformation structural components **200** may be controlled firstly to drive the driven module **100** to rotate to a specified direction, and then the second field deformation structural components **400** may be controlled to drive the driven module **100** to stretch out of the shell for work. Of course, the second field deformation structural components **400** may be controlled firstly to drive the driven module **100** to stretch out of the shell. Further, the first field deformation structural components **200** are controlled to drive the driven module **100** to rotate to a specified direction for work.

In an optional solution, the first field deformation structural components **200** and the second field deformation structural components **400** may be controlled at the same time. Specifically, in the process that the first field deformation structural components **200** drive the driven module **100** to rotate, the second field deformation structural components **400** drive the driven module **100** to stretch out of the shell, so that the two driving effects of the driving mechanism on the driven module **100** are performed simultaneously, and finally the working efficiency of the driving mechanism is improved.

The first field deformation structural components **200** and the second field deformation structural components **400** may be arranged in the avoidance space in various manners. For example, the first field deformation structural components **200** and the second field deformation structural components **400** may be distributed at intervals in the telescoping direction of the driven module **100**, or the first field deformation structural components **200** and the second field deformation structural components **400** may also be distributed at intervals in the rotating direction of the driven module **100**. Of course, in an optional solution, the first field deformation structural components **200** and the second field deformation structural components **400** may be distributed at intervals in the rotating direction of the driven module **100**. Compared with other arrangement manners, this manner is more favor-

able for the mounting of the first field deformation structural components **200** and the second field deformation structural components **400**.

Meanwhile, in this case, the first field deformation structural components **200** may be arranged in rows in the avoidance space. Similarly, the second field deformation structural components **400** may also be arranged in rows in the avoidance space. Of course, in an optional solution, the first field deformation structural components **200** and the second field deformation structural components **400** may be arranged in a crossed manner in the avoidance space. In this manner, one row of second field deformation structural components **400** may be arranged at intervals between any two rows of first field deformation structural components **200**, and each row of first field deformation structural components **200** and each row of second field deformation structural components **400** are arranged along the telescoping direction of the driven module **100**. This distribution manner can fulfill the aim of balancing driving.

Of course, the driving mechanism may further include a second elastic portion. Specifically, the second elastic portion is arranged between the driven module **100** and the second field deformation structural components **400**. The second elastic portion can enlarge the traveling wave formed by the second field deformation structural components **400**, so that the driving effect of the second field deformation structural components **400** on the driving module **100** is more obvious.

In the embodiments disclosed by the present disclosure, the avoidance space may be formed on the middle frame **500** of the shell, or may also be formed at other positions of the shell. Generally, in order to make the appearance performance of the electronic device better, the avoidance space may be formed on the middle frame **500** of the shell. Specifically, the avoidance space may be an avoidance hole. Of course, in order to facilitate the mounting of the first field deformation structural components **200**, the avoidance space may be a groove, and the first field deformation structural components **200** may be paved on an inner wall of the groove.

In order to make the rear cover **600** of the shell not affect the movement of the driven module **100**, in the embodiments of the present disclosure, the middle frame **500** may be provided with a first groove **510**, the rear cover **600** may be provided with a second groove **610**, and the first groove **510** and the second groove **610** form an avoidance space. In this case, the compact structure of the electronic device can be ensured, and the mounting of the first field deformation structural components **200** can be facilitated.

Of course, the first field deformation structural components **200** may also be arranged on an inner wall of the second groove **610**, so that the first field deformation structural components **200** can drive the driven module **100** to move at a plurality of positions of the driven module **100**, so that the driving effect of the first field deformation structural components **200** on the driven module **100** is improved. Meanwhile, in order to improve the movement stability of the driven module **100**, the first field deformation structural component **200** on the first groove **510** and the first field deformation structural component **200** on the second groove **610** may be arranged oppositely, so that the driven module **100** is subjected to a balanced driving force at each position, and the driven module **100** can move stably.

Generally, in order to facilitate electrical connection of the first field deformation structural components **200**, the electronic device may further include a circuit layer **700**, wherein the circuit layer **700** may be arranged in the

avoidance space, the first field deformation structural components **200** may be arranged on the circuit layer **700** and are electrically connected to the circuit layer **700**, so that the energizing current of each of the first field deformation structural components **200** can be controlled conveniently.

Of course, the middle frame **500** is generally a metal piece, so in order to prevent the short circuit of the middle frame **500** and the circuit layer **700**, the electronic device may further include an insulating layer **800**. In the specific assembling process, the insulating layer **800** may be paved on the avoidance space, and the circuit layer **700** may be paved on the insulating layer **800**, so that the middle frame **500** is separated from the circuit layer **700** through the insulating layer **800**, and the short circuit of the middle frame **500** and the circuit layer **700** is avoided. Meanwhile, in order to ensure the connection reliability of each layer, each layer may be stacked together in an adhesion manner.

In the electronic device disclosed by the embodiments of the present disclosure, the first field deformation structural components **200** and the second field deformation structural components **400** may realize driving functions in various manners, for example, the first field deformation structural components **200** and the second field deformation structural components **400** may all be electrostrictive pieces. The electrostrictive pieces are formed by electrostrictive materials, and the electrostrictive materials may be one of a piezoelectric single crystal, a piezoelectric polycrystal, a piezoelectric polymer or a piezoelectric composite material.

The specific working manners of the electrostrictive pieces are: when an electric field is applied to the surface of the piezoelectric material, an electric dipole moment will be lengthened due to the action of the electric field, and the piezoelectric material will stretch along the direction of the electric field to resist the change, so that the aim of driving the driven module **100** is fulfilled. Of course, the electrostrictive pieces may be made of various materials. In an optical solution, the electrostrictive pieces may be shape memory alloy pieces. Compared with the electrostrictive pieces made of other materials, the shape memory alloy has longer service life.

Of course, the first field deformation structural components **200** and the second field deformation structural components **400** each may include electromagnetic coils and magnetostrictive pieces, wherein the electromagnetic coils and the magnetostrictive pieces are arranged in the avoidance space, the magnetostrictive pieces are connected to the driven module **100**, the electromagnetic coils can generate a magnetic field during energization, and the magnetostrictive pieces can deform in the magnetization direction after being magnetized in the magnetic field. Specifically, the magnetostrictive pieces may be one of an alloy telescoping material, a piezoelectric ceramic material or a rare earth intermetallic compound magnetostrictive material. In the specific working process, when the electromagnetic coils are in a power-up state, the magnetostrictive pieces are magnetized through the magnetic field at the periphery of the electromagnetic coils, so that the magnetostrictive pieces deform, and the driven module **100** is finally driven to move.

The first field deformation structural components **200** and the second field deformation structural components **400** may all photostrictive pieces, the materials of the photostrictive pieces are photostrictive materials, and the photostrictive materials can deform under the irradiation of light beams, so that the photostrictive pieces can drive the driven module **100** to move through deformation. Of course, the photostrictive materials may be a ferroelectric material, photo-sensitive perovskite and the like.

The electronic device disclosed in the embodiments of the present disclosure may be a smart phone, a tablet computer, an e-book reader, a wearable device and the like. Of course, the electronic device may also be other devices, and the embodiments of the present disclosure do not limit the specific types of the electronic device.

The foregoing embodiments of the present disclosure focus on describing differences between the embodiments, and different optimization features of the embodiments may be combined to form better embodiments provided that they are not contradictory. Considering brevity, details are not described herein again.

The foregoing descriptions are merely embodiments of the present disclosure, but are not intended to limit the present disclosure. For a person skilled in the art, various modifications and changes may be made to the present disclosure. Any modification, equivalent replacement, or improvement made without departing from the spirit and principle of the present disclosure shall fall within the scope of the claims of the present disclosure.

The invention claimed is:

1. An electronic device, comprising a shell, a driving mechanism and a driven module, wherein the shell is provided with an avoidance space communicating with an inner cavity of the shell; the driving mechanism comprises at least two first field deformation structural components; the at least two first field deformation structural components are arranged in the avoidance space; wherein every two adjacent first field deformation structural components among the at least two first field deformation structural components are distributed at intervals and have opposite energizing currents; the driven module is connected to the at least two first field deformation structural components; and when the at least two first field deformation structural components are in a power-up state, each of the at least two first field deformation structural components drives the driven module to rotate through deformation.

2. The electronic device according to claim 1, wherein the first driving mechanism further comprises a first elastic portion, and the first elastic position is arranged between the driven module and the at least two first field deformation structural components.

3. The electronic device according to claim 1, wherein the driving mechanism further comprises at least two second field deformation structural components; the at least two second field deformation structural components are arranged in the avoidance space; wherein every two adjacent second field deformation structural components among the at least two second field deformation structural components are distributed at intervals and have opposite energizing currents; and when the at least two second field deformation structural components are in a power-up state, each of the at least two second field deformation structural components

drives the driven module through deformation to retract in the shell through the avoidance space or at least partially stretch out of the shell.

4. The electronic device according to claim 3, wherein the at least two first field deformation structural components and the at least two second field deformation structural components are distributed at intervals in a rotating direction of the driven module.

5. The electronic device according to claim 3, wherein the driving mechanism further comprises a second elastic portion, and the second elastic portion is arranged between the driven module and the at least two second field deformation structural components.

6. The electronic device according to claim 1, wherein the shell comprises a middle frame and a rear cover, the middle frame is connected to the rear cover, the middle frame is provided with a first groove, the rear cover is provided with a second groove, the first groove and the second groove form the avoidance space, and at least one of the at least two first field deformation structural components is arranged on an inner wall of the first groove.

7. The electronic device according to claim 6, wherein at least another one of the at least two first field deformation structural components is arranged on an inner wall of the second groove; and the at least one of the at least two first field deformation structural components on the first groove and the at least another one of the at least two first field deformation structural components on the second groove are arranged opposite to each other.

8. The electronic device according to claim 6, further comprising a circuit layer and an insulating layer, wherein the insulating layer is paved on the avoidance space, the circuit layer is paved on the insulating layer, the at least two first field deformation structural components are arranged on the circuit layer and are electrically connected to the circuit layer.

9. The electronic device according to claim 3, wherein the at least two first field deformation structural components and the at least two second field deformation structural components are electrostrictive pieces.

10. The electronic device according to claim 3, wherein the at least two first field deformation structural components and the at least two second field deformation structural components are photostrictive pieces.

11. The electronic device according to claim 3, wherein the at least two first field deformation structural components and the at least two second field deformation structural components each comprises electromagnetic coils and magnetostrictive pieces; the electromagnetic coils and the magnetostrictive pieces are arranged in the avoidance space; the magnetostrictive pieces are connected to the driven module; and when the electromagnetic coils are in a power-up state, the magnetostrictive pieces deform to drive the driven module to rotate.

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